

VELORA MOBITECH

Team ID - 2699

Employee Transport Optimization System

Solving the Vehicle Routing Problem for Corporate Fleets

THE PROBLEM

STATUS QUO

1. Employees individually book Ola/Uber to office
2. Rs. 50–80/km per person — highly expensive
3. Uncoordinated arrivals — no scheduling
4. Redundant trips across overlapping routes
5. No shared fleet — massive waste of resources



VELORA SOLUTION

1. Shared corporate fleet — fewer vehicles, more riders
2. Rs. 10–18/km fleet rate — 81–95% cost reduction
3. Algorithmic routing — optimal pickup sequences
4. Automated scheduling — everyone on time
5. Excel upload → optimised routes in seconds

SYSTEM ARCHITECTURE

FRONTEND

React + Vite

- File Upload (.xlsx)
- Leaflet Map Display
- Route Visualization
- Assignment Dashboard

Hosted: Vercel

BACKEND

Node.js + Express

- API Gateway
- CORS Middleware
- Multer File Upload
- Axios → Python Relay

Hosted: Render

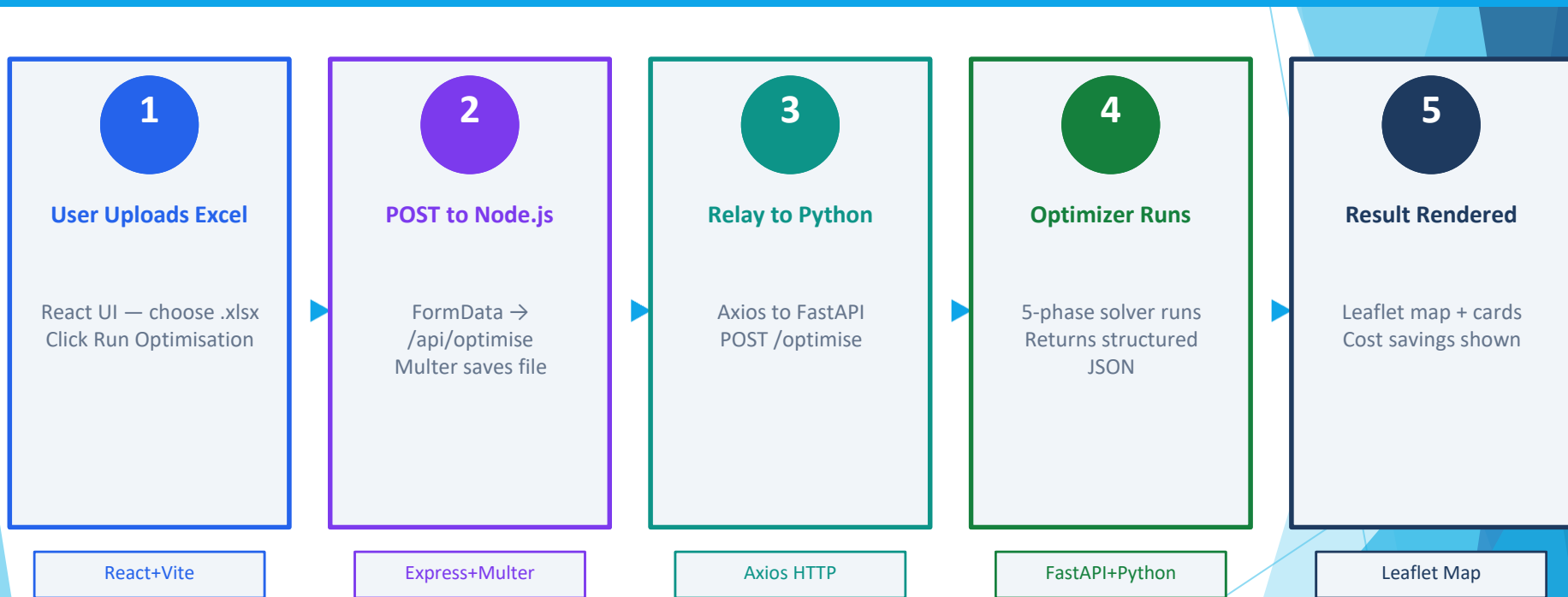
OPTIMIZER

Python + FastAPI

- K-Means Clustering
- Greedy Insert (3-pass)
- 2-Opt Improvement
- JSON Result Output

Hosted: Render

REQUEST LIFECYCLE FLOW



THE OPTIMIZATION ENGINE

5-Phase Solver: K-Means Clustering → Greedy Insert → Repair → Relaxed Insert → 2-Opt

P1

Cluster + Sort

- K-Means clusters employees by GPS
- Tightest deadline sorted first
- Natural ride-share groupings formed

P2

Greedy Insert (Hard)

- Try each employee in every slot
- Hard reject on pref violations
- Score = $0.6 \times \text{cost} + 0.4 \times \text{time}$

P3

Repair Phase

- Find unassigned after Phase 2
- Swap with highest-slack employee
- Re-place displaced employee safely

P4

Relaxed Insert

- Retry with prefs relaxed
- Violations penalised +0.30/rider
- Force-assign remaining → flag infeasible

P5

2-Opt Improve

- Reverse route sub-sequences
- Keep if distance drops
- Repeat until no improvement

Objective: score = (time_weight × norm_arrival) + (cost_weight × norm_cost) + violation_penalty

SOLVER FLOW: SETUP & PHASE 1 — CLUSTER + SORT

GRAPH SETUP



Vehicles

Nodes 0 → V-1



Employees

Nodes V → V+E-1



Factory

Node V+E (last)

Haversine Distance Matrix

Pairwise GPS km computed once for all nodes. O(1) lookup during solving.

PHASE 1: K-MEANS CLUSTER + SORT BY DEADLINE

Read all employee GPS coordinates (lat, lng)

Run K-Means with k = number of vehicles
→ Each cluster = natural vehicle grouping

Within each cluster: sort employees by deadline
(latest_drop + priority_delay_allowance)
Tightest deadline → served first

Output: ordered employee list ready
for greedy insertion in Phase 2

SOLVER FLOW: PHASE 2 — GREEDY INSERT (HARD CONSTRAINTS)

For each employee (deadline order): try every position in every vehicle. Hard reject if prefs violated. Pick best-scoring.

Take next employee from sorted list

For each vehicle V in fleet:

For each insertion position P in V's route:

Check HARD constraints:

- Sharing pref: riders $\leq$ limit?
- Vehicle type: standard / premium?

HARD FAIL $\rightarrow$ skip slot entirely (no penalty)

PASS $\rightarrow$ run simulate() for time check

$\rightarrow$ score = cost_weight $\times$ norm_cost + time_weight $\times$ norm_time

Assign to lowest-score valid slot

simulate() — Feasibility Engine

1. Start from vehicle's available time
2. Drive to each pickup stop in order
3. Add travel time (Haversine km $\div$ speed)
4. Wait if arrives before earliest_pickup
5. After all pickups $\rightarrow$ drive to factory
6. HARD: any drop $>$ deadline $\rightarrow$ False
7. SOFT: return actual time (infeasibility pass)

score = cost_weight $\times$ norm_cost + time_weight $\times$ norm_arrival
[both normalised 0–1]

SOLVER FLOW: PHASES 3, 4 & 5 — REPAIR → RELAX → 2-OPT

PHASE 3: REPAIR

Identify all unassigned
after Phase 2

Find assigned employee
with most schedule slack

Temporarily remove
that employee from slot

Insert unassigned into
vacated slot (Hard prefs
still enforced)

Find new slot elsewhere
for displaced employee

Both OK → commit swap
Either fails → revert

PHASE 4: RELAXED INSERT

Employees still unassigned
after repair → enter here

Re-try insertion with
prefs NOW RELAXED

Penalty added to score:
+0.30/excess rider
+0.10 wrong veh type

Solver picks LEAST-violating
option available

Still unassigned:
Force-assign → flag
is infeasible: true
Record delay_min

GUARANTEE: all violations
are mathematically
unavoidable given fleet

PHASE 5: 2-OPT

All employees assigned.
Begin route improvement.

Per vehicle: select two
positions i and j
in the route

Reverse sub-sequence
between positions i, j

Re-run simulate() —
does total distance
decrease?

YES → keep reversal
NO → revert to original

Repeat until full pass
has zero improvements.
Route locally optimal.

CONSTRAINT HANDLING

HARD CONSTRAINTS

(Never violated — hard reject in solver)

Every employee assigned to exactly 1 vehicle

No vehicle exceeds seating capacity

Pickup $\geq$ employee earliest_pickup time

Drop $\leq$ latest_drop + priority_delay_allowance

SOFT CONSTRAINTS

(Best-effort — penalised only in relaxed pass)

Sharing pref (single/double/triple) +0.30 per excess rider

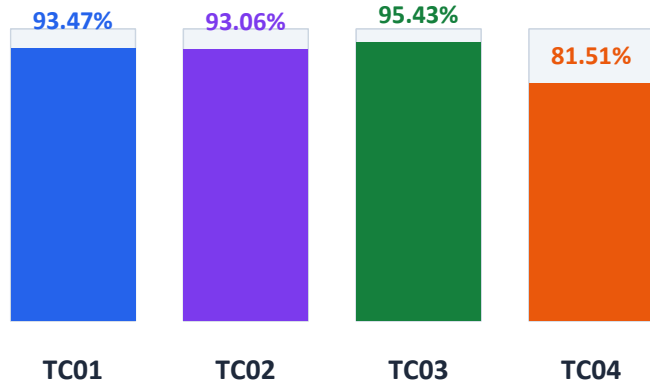
Vehicle type pref (standard/premium) +0.10 wrong type

Minimize fleet completion time + total fleet distance

TEST CASE RESULTS SUMMARY

Test Case	Employees	Vehicles	Baseline Cost	Fleet Cost	Savings %	Time Saved
TC01	8	3	Rs. 3,200	Rs. 209	93.47%	56 min
TC02	12	5	Rs. 4,945	Rs. 343	93.06%	52 min
TC03	15	6	Rs. 6,450	Rs. 295	95.43%	48 min
TC04	10	4	Rs. 4,420	Rs. 817	81.51%	54 min

Cost Savings per Test Case



100%

Assignment Rate
across all TCs

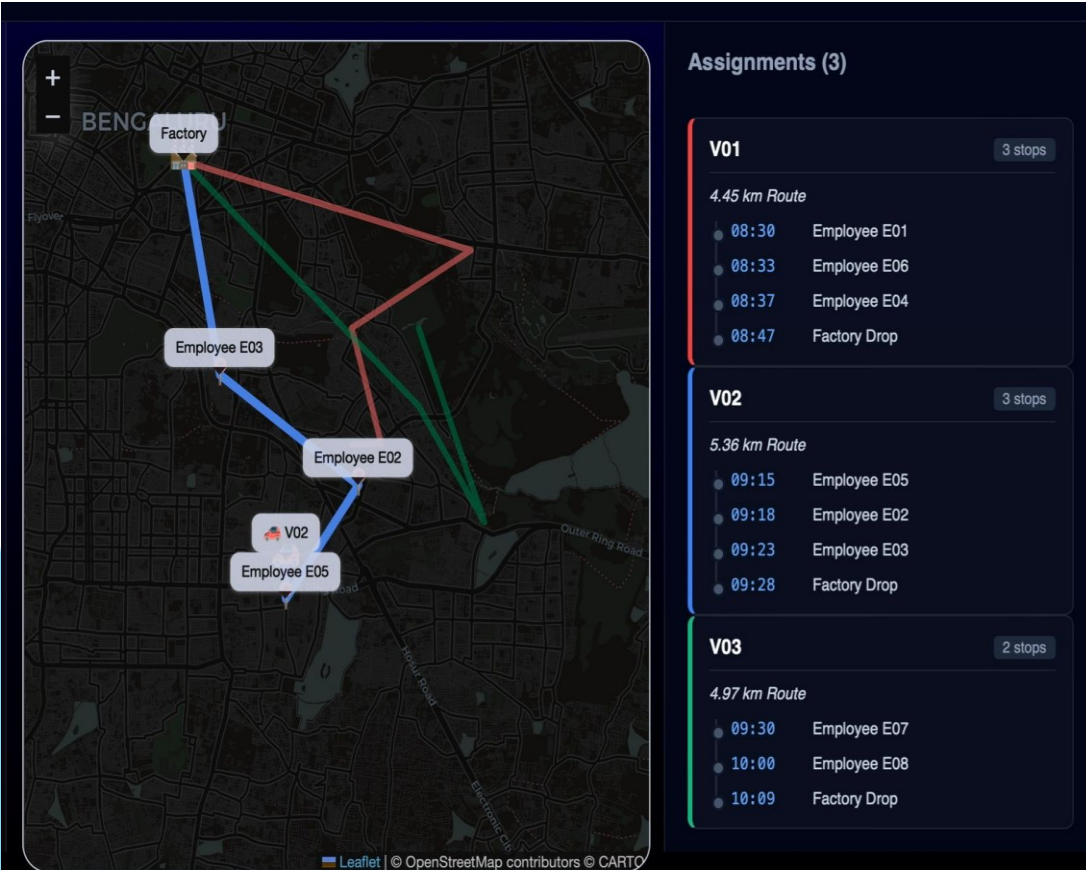
50 min

Avg Time Saved
vs Ola/Uber

~90%

Average
Cost Reduction

TC01 — DETAILED RESULTS (8 Employees, 3 Vehicles)



Rs.3,200

Baseline Cost

Rs.209

Fleet Cost

93.47%

Cost Saved

10:09

Fleet Completes

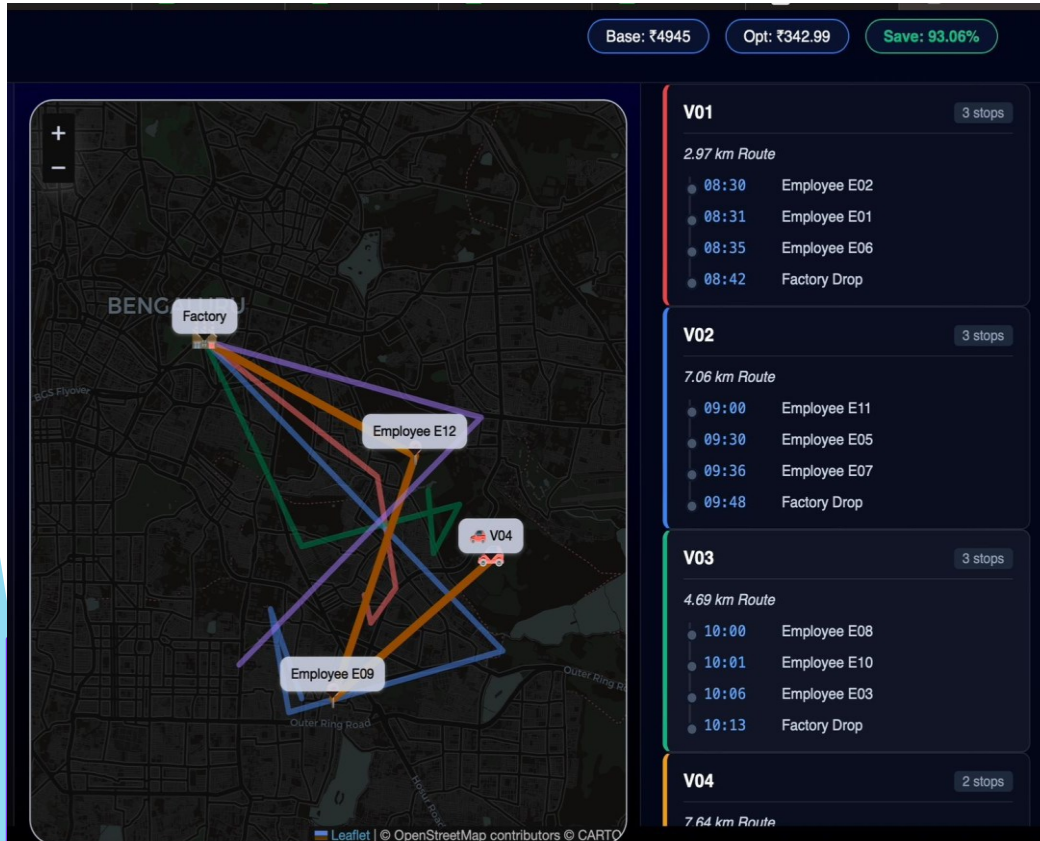
56 min

Time Saved

8 / 8

Assignment

TC02 — DETAILED RESULTS (12 Employees, 5 Vehicles)



Rs.4,945

Baseline Cost

Rs.343

Fleet Cost

93.06%

Cost Saved

10:13

Fleet Completes

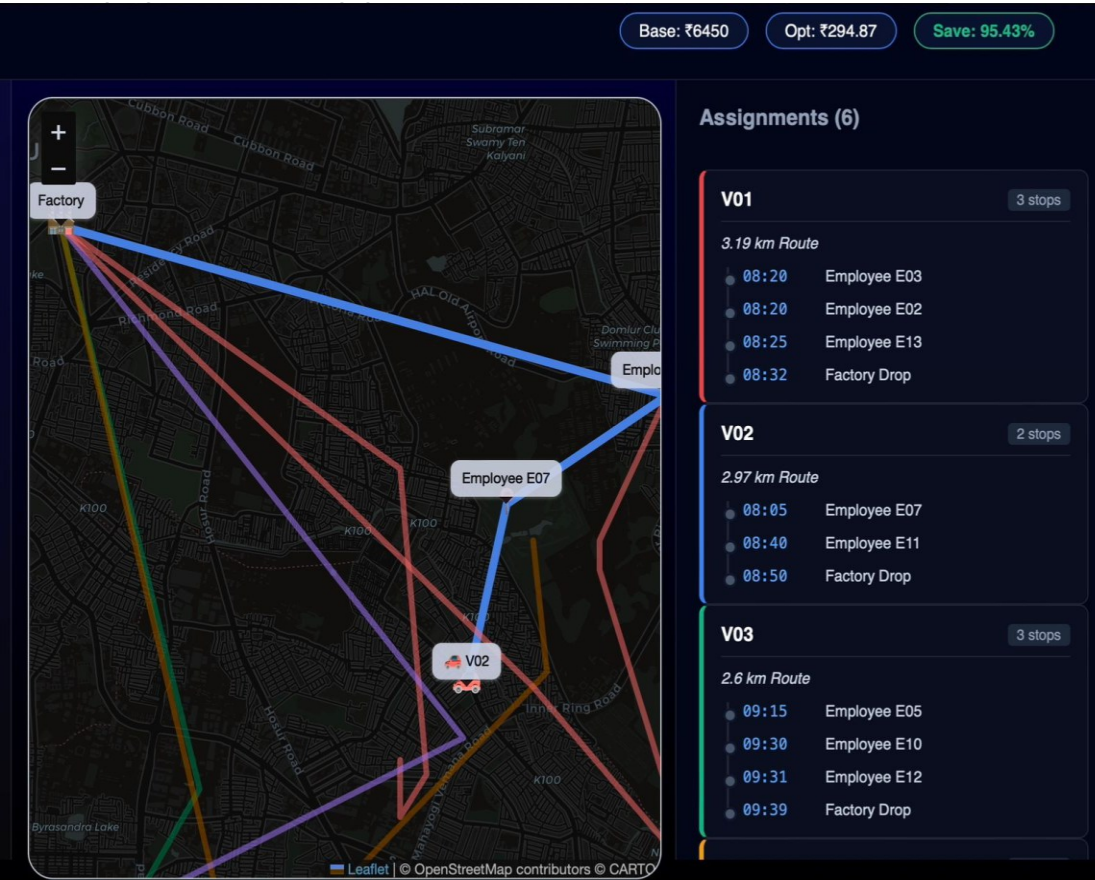
52 min

Time Saved

12/12

Assignment

TC03 — DETAILED RESULTS (15 Employees, 6 Vehicles)



Rs.6,450

Baseline Cost

Rs.295

Fleet Cost

95.43%

Cost Saved

10:17

Fleet Completes

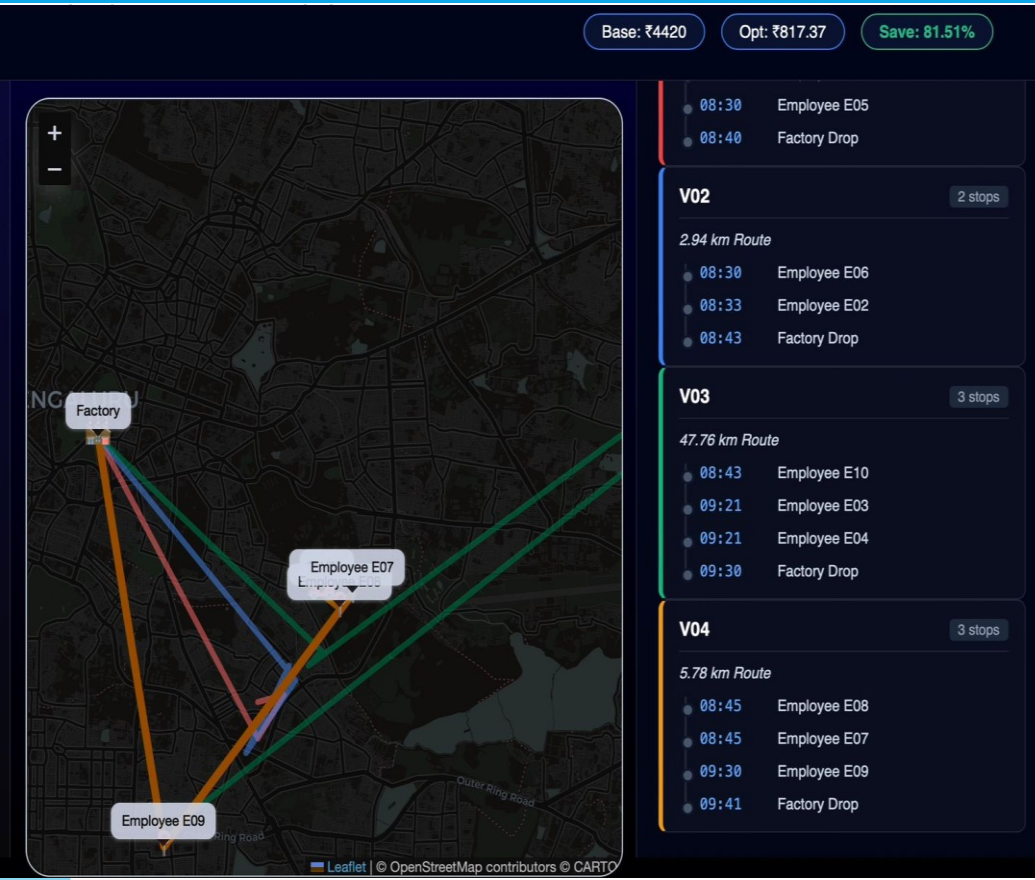
48 min

Time Saved

15/15

Assignment

TC04 — DETAILED RESULTS (10 Employees, 4 Vehicles — Infeasible)



E10 — ORIGIN

Deadline: 08:45

Actual Drop: 09:30

Delay: 45 min

23.9 km from office. Min travel=47.8 min at 30 kmph.
Window=40 min. Mathematically impossible.

E03 — CASCADE

Deadline: 09:00

Actual Drop: 09:30

Delay: 30 min

Same vehicle as E10. E10's distant pickup adds unavoidable detour to entire route.

E04 — CASCADE

Deadline: 09:05

Actual Drop: 09:30

Delay: 25 min

Same vehicle as E03 & E10. Fleet has insufficient vehicles to isolate E10 alone.

TC04 — DETAILED RESULTS (10 Employees, 4 Vehicles — Infeasible)

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Base: ₹4420

Opt: ₹817.37

Save: 81.51%

Employee Preferences and Cost Comparison

Employee ID	Vehicle Preference	Sharing Preference	Assigned Vehicle	Actual Sharing	Vehicle Pref OK	Sharing Pref OK	Baseline Cost	Bas
E01	premium	single	premium	2	✓	✗	₹420	40
E02	premium	single	premium	2	✓	✗	₹425	42
E03	premium	single	normal	3	✗	✗	₹430	43
E04	premium	single	normal	3	✗	✗	₹435	44
E05	any	double	premium	2	✓	✓	₹400	50
E06	any	double	premium	2	✓	✓	₹395	50
E07	normal	triple	normal	3	✓	✓	₹380	58
E08	normal	triple	normal	3	✓	✓	₹385	58
E09	any	triple	normal	3	✓	✓	₹450	65
E10	premium	single	normal	3	✗	✗	₹700	35

Employee E09

HACK TEST CASES — WHY THESE 3 EDGE CASES?

We designed 3 hack test cases to probe the exact algorithmic features that make our solver robust — and that a weak solver would fail on.

HackTest_PreferenceOverload

WHY this catches weak solvers:

A naive solver that just assigns everyone greedily with no preference logic will either crash, leave employees unassigned, or give premium seats to the wrong priority employees.

HOW our solver handles it:

Our two-pass design is built exactly for this. Pass 1 tries all preferences strictly. Pass 2 relaxes only when fleet is provably insufficient. Priority-1 employees get the 2 premium seats first because we sort by deadline before assignment.

10/10 assigned • 96.06% savings • Violations correctly unavoidable

HackTest_ZigzagRouting

WHY this catches weak solvers:

A solver that assigns correctly but has no route improvement will zigzag north-south repeatedly — wasting ~29km extra distance per vehicle. Tests route QUALITY not just assignment correctness.

HOW our solver handles it:

KMeans clustering naturally groups north employees together and south employees together. 2-opt then eliminates any backtracking within each cluster. A solver without either of these features will waste huge distance.

8/8 assigned • 86.63% savings • 0 violations • Clusters geographically tight

HackTest_PriorityDeadline

WHY this catches weak solvers:

A greedy nearest-first solver will pick up priority-5 employees first because they are closer. This leaves priority-1 employees — who are far away and have tight 5-min tolerance — unable to meet their deadline.

HOW our solver handles it:

Our Phase 1 sorts ALL employees by tightest deadline before any insertion. Priority-1 employees are processed first and get first pick of vehicle slots. Priority-5 employees — who have huge windows — get whatever remains, and they still easily make their deadline.

6/6 assigned • 79.03% savings • 0 missed deadlines • All P1 on time

HACK TC1 — PREFERENCE OVERLOAD

SCENARIO — EDGE CASE TARGETED

Targets: Simultaneous sharing + vehicle pref enforcement under tight fleet

- 10 employees — ALL want single-rider
- 5 want premium + single
- 5 want normal + single
- Only 1 premium vehicle (capacity 2)
- 2 normal vehicles (capacity 4 each)
- Fleet has ZERO solo slots
- Weak solver: may leave employees unassigned
- Weak solver: gives premium to wrong-priority employees

OUR SOLUTION RESULT

Vehicle	Riders	Dist	Cost	Notes
V01 (premium)	2	1.30 km	Rs.12.99	Top-priority premium
V02 (normal)	4	6.19 km	Rs.74.30	Normal-pref employees
V03 (normal)	4	5.65 km	Rs.79.17	Remaining, relaxed-pass
TOTAL	10	13.14 km	Rs.166.46	vs Rs.4,725 baseline

10/10

Assigned

96.06%

Savings

3

Pref violations
(unavoidable)

10

Sharing violations
(0 solo slots)

Priority-1 employees received the 2 premium seats first (Phase 1 deadline sort).

HACK TC2 — ZIGZAG ROUTING

SCENARIO — EDGE CASE TARGETED

Targets: K-Means clustering + 2-opt on geographic spread

- 8 employees in 2 tight geographic clusters
- 4 employees north of factory
- 4 employees south of factory
- 3 vehicles available
- Without clustering: zigzags $N \leftrightarrow S$ — ~29 km wasted per vehicle
- Without 2-opt: backtracking within each cluster
- Tests route quality, not just assignment correctness
- Weak solver: assigns everyone but wastes huge distance

OUR SOLUTION RESULT

Vehicle	Riders	Dist	Cost	Notes
V01	3	19.35 km	Rs.232.22	N+S mix (2-opt opt.)
V02	2	8.21 km	Rs.98.54	North cluster only
V03	3	8.32 km	Rs.116.41	South cluster only
TOTAL	8	35.88 km	Rs.447.17	vs Rs.3,345 baseline

8/8

Assigned

86.63%

Savings

0

Sharing violations

0

Type violations

K-Means kept clusters geographically tight. 2-opt eliminated backtracking.

HACK TC3 — PRIORITY DEADLINE ORDERING

SCENARIO — EDGE CASE TARGETED

Targets: Priority-first deadline ordering vs nearest-first greedy

- 3 Priority-1 employees, 10–14 km from office
5-minute deadline tolerance only
- 3 Priority-5 employees, 1–2 km from office
30-minute deadline tolerance
- Weak solver: serves P5 first → P1 misses deadline
- Our solver: sorts by TIGHTEST DEADLINE first
- Phase 1 guarantees P1 served before P5
regardless of distance

OUR SOLUTION RESULT

Vehicle	Riders	Dist	Cost	Employees
V01	3	31.89 km	Rs.382.69	E03(P1),E01(P1),E06(P5)
V02	3	11.76 km	Rs.164.66	E05(P5),E04(P5),E02(P1)
TOTAL	6	43.65 km	Rs.547.35	vs Rs.2,610 baseline

All Priority-1 deadlines met:

E01	P1	Drop 09:05	Deadline 09:35
E02	P1	Drop 08:43	Deadline 09:35
E03	P1	Drop 09:05	Deadline 09:35
E04	P5	Drop 08:43	Deadline 11:30
E05	P5	Drop 08:43	Deadline 11:30
E06	P5	Drop 09:05	Deadline 11:30

6/6 Assigned

79.03% Savings

0 Missed deadlines

KEY ACHIEVEMENTS

Velora Mobitech — End-to-End VRP Solution

100% Assignment Rate

All employees assigned
across all 4 test cases

81–95% Cost Savings

vs individual Ola/Uber
across all scenarios

48–56 Min Earlier

Fleet completion vs
individual cab baseline

Hard Constraint Guarantee

Pass 1 enforces all prefs —
violations only when fleet
under-provisioned

Explicit Infeasibility

Flagged with exact delay_min
— never silently dropped

2-Opt Optimization

Route improvement reduces
total fleet distance